Gradient Descent Optimization

Gradient Descent is an optimization algorithm used to minimize a loss function.

Learning Rate determines the step size at each iteration while moving toward a minimum.

Loss Function measures how far estimated values are from actual values.

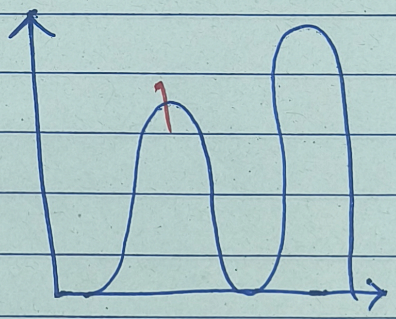
Global Minimum represents the state with the lowest possible loss.

Simple Hill Climbing Algorithm (Local search, Greedy approach, No back track)

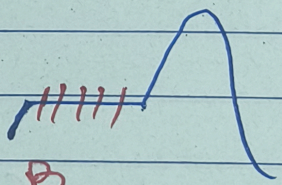
1. Evaluate the initial state.
2. Loop ~~until~~ until a ~~sol~~ solution is found or there are no ~~problems~~ operators ~~left~~ left.
   - Select & apply a new operator.
   - Evaluate the new state.
     if goal then quit.
     if better than the current state then it is now current state.

Limitations

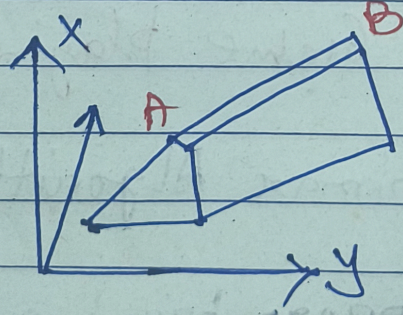
1) Local Maxima



2) Plateau/Flat Maxima



3) Ridge



(Does not change direction)